Jenna Kim, Boston, MA JieHyun.Kim001@umb.edu +1 857-316-6290

Computational researcher specializing in deep learning based medical image segmentation. My work focuses on developing clinically interpretable AI systems for cardiovascular imaging, bridging methodological advances with real-world biomedical applications.

Education

2026	PhD in Computer Science, University of Massachusetts Boston *Streamlined Biomedical Image Processing Pipelines* Advisor: Daniel Haehn Committee: Jae W. Song, Nurit Haspel, Tales Imbiriba	Boston, MA
2019	Master of Science in Business Analytics, University of Massachusetts Boston	Boston, MA
2017	Bachelor of Science in Business Administration	Kearney, NE

Experience

Summer 2025 — AI Developer, myotwin — Göttingen, Germany

• Built a preprocessing pipeline for cardiac organoid image analysis, processing datasets of over 6,000 frames using UNet based ROI segmentation.
• Optimized motion tracking using Lucas-Kanade optical flow, reducing processing time from hours to under 1 minute for contraction-phase analysis.
• Developed end-to-end workflow integrating segmentation, signal processing, motion tracking, and velocity calculation.
• Implemented ZenML for pipeline orchestration to support scalable analysis and future automation.

2025-2026 — Research Assistant — Boston, MA

• Significantly contributed to securing $20,000 internal grant for *"Automatic Plaque Segmentation in Carotid Arteries with Uncertainty Quantification"* (July 2025 – December 2026).

Publications

First Author Publications

2026 **Automated Analysis of Calcium Transients in Engineered Cardiac Microtissues**
• Developed an automated pipeline for spatial analysis of calcium-driven motion in engineered cardiac tissue imaging, integrating segmentation, event detection, and optical flow-based motion tracking.
• The paper was submitted to the *Medical Image Computing and Computer Assisted Intervention (MICCAI).*

2025 **Automatic Segmentation of Calcified Plaque in Carotid Arteries**
• Developed an automated method to segment small calcified plaques in carotid arteries using a two-step approach: first segmenting the carotid artery to define the region of interest, then detecting plaques within that region.
• The paper was accepted at the *IEEE International Symposium on Biomedical Imaging (ISBI).*

2023 **Streamlined Carotid Artery Calcification Labeling for CTA Scans**
• Compared existing plaque annotation tools with a newly developed manual labeling tool (CACTAS Tool) for calcified plaque annotation in CTA scans.
• The paper was accepted at the *Society of Cardiovascular Computed Tomography (SCCT).*

Co-Author Publications

2026 Sakai Yu, JieHyun Kim et al. “**Explainable machine-learning model to classify culprit calcified carotid plaque in embolic stroke of undetermined source**” Journal of Neuroimaging (2026)

2025 Sakai Yu, JieHyun Kim et al. "**Abstract WMP65: Machine-learning Approach To Classify Vulnerable Calcified Plaque In Embolic Stroke Of Undetermined Source.**" *Stroke (2025).*

2024 Bennie Bendiksen, Nana Lin, Jiehyun Kim, Funda Durupinar. "**Assessing human reactions in a virtual crowd based on crowd disposition, perceived agency, and user traits.**" *ACM Transactions on Applied Perception* 21.3 (2024): 1-21.

2022 Durupinar, Funda, and Jiehyun Kim. "**Facial emotion recognition of virtual humans with different genders, races, and ages.**" *ACM Symposium on Applied Perception 2022.*

Presentations

2026	UMB Departmental Graduate Students' Symposium: Streamlined Biomedical Image Processing Pipelines
2025	Invited speaker at Biosciences & Computational Sciences Graduate Program Seminar @UMB
2025	UMB Machine Psychology Group meeting: Calcium Transit Analysis in Engineered Cardiac Tissue
2025	UMB Departmental Graduate Students' Symposium: Automatic Segmentation of Calcified Plaque in Carotid Arteries
2024	UMB Machine Psychology Group meeting: CACTAS
2024	UMB Departmental Graduate Students' Symposium: Automatic Labeling of Calcific Plaque in Carotid Arteries
2023	UMB Machine Psychology Group meeting: Kaggle HubMAP Organ Segmentation

Mentoring

2025	Table Lunch Session Discussion Leader, ACM MAGICC Conference • Facilitated informal discussions on graduate school and research career.
2024-2025	Daniel Gross, University of Massachusetts Boston, IDS Sloan Program • Mentored on research skills and preparation for graduate study.
2024	Dhruv Shah, University of Massachusetts Boston • Provided guidance on research projects and technical development.
2024	Project Lead, Brainhack @UMB • Introduced and supported a medical imaging project during an interdisciplinary hackathon.

Skills

Programming	Python, TensorFlow, Keras, Deep Learning, Computer Vision
Medical Imaging	CTA, MRI, Segmentation, UNet, nnUNet, Image Processing
Tools	Git, LaTeX, ZenML

References

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Dr. Jae W. Song
Associate Professor, Department of Radiology
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